Non-linear simultaneous equations

1 Solve the following systems of equations:

a $y = x^2$
$y = 4x$

b $y = 2x$
$y = x^2 + 6x$

c $x^2 + y = 2$
$y = x - 4$

d $y = x^2 - 16$
$y = x - 16$

e $x^2 + y = 10$
$2x - y = 5$

f $x^2 + y = 5$
$-2x + y = 5$

g $y = x^2 + 2$
$y = 4 - x$

h $y = (x - 2)^2$
$y = 8 - x$

i $y = x^2 - 9x - 6$
$y = x + 5$

j $y = 2x^2 - 8x + 10$
$y = 4x - 8$

k $y = -2x^2 - 3x + 7$
$y = 4x - 8$

l $y = 2x^2 + 4x + 2$
$y = 2x + 6$

2 Approximate the solutions of the following systems of equations by graphing the functions on a number plane and approximating their points of intersection. Round your values to one decimal place.

a $y = x^2 - 4$
$y = x - 1$

b $y = x + 4$
$y = 8 - x^2$

Applications

3 The sum of two numbers is 60. The larger number equals the smaller number squared plus 4.

a Set up two equations by letting x and y be the two numbers, where y is the larger of the two numbers.

b Find all the solutions for x and y. Write your answers as ordered pairs.

4 In business, supply and demand functions [illegible]ays of expressing the supply or demand of a commodity as a function of its unit price. Market equilibrium is the term that describes when the supply is equal to the demand. Consider the supply and demand functions for a new book bag below, where y is in dollars and x is in thousands of units supplied:

- Supply function: $y = 2x^2$
- Demand function: $y = 4x + 6$

a Solve the equations for x.

b Which of these values of x must the equilibrium point be at? Explain your answer.

c Find the value of y for this value of x.

d Complete the following statement:

The equilibrium quantity occurs when $\boxed{}$ thousand units are being produced, which has a corresponding price of $\boxed{}$ dollars per unit.